

Formula One® e.Report Engine

Application Data Reporting

*10 Quick Steps to Building Powerful Reports
By Mapping Java Objects To a Relational Data Table Model*



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Last updated 6/16/03.

Application Data Reporting

► Overview

The traditional data reporting model accesses static data residing in persistent storage such as databases. Working with the old data reporting model using new software development techniques can be difficult. A new application data reporting model available in the Formula One e.Report Engine and Designer solves these difficulties.

Application Data Reporting refers to reporting on data from Java application objects rather than from a database.

Advantages of application data reporting include:

► Efficiency

Application data reporting is much more efficient than database reporting in terms of computing resources. You often have to query a database repeatedly for static data. You query application data once and access it repeatedly.

► Support for Distributed Applications

Many of today's applications use a distributed architecture and data source. Accessing multiple data sources to create reports is difficult, if not impossible, using traditional data reporting.

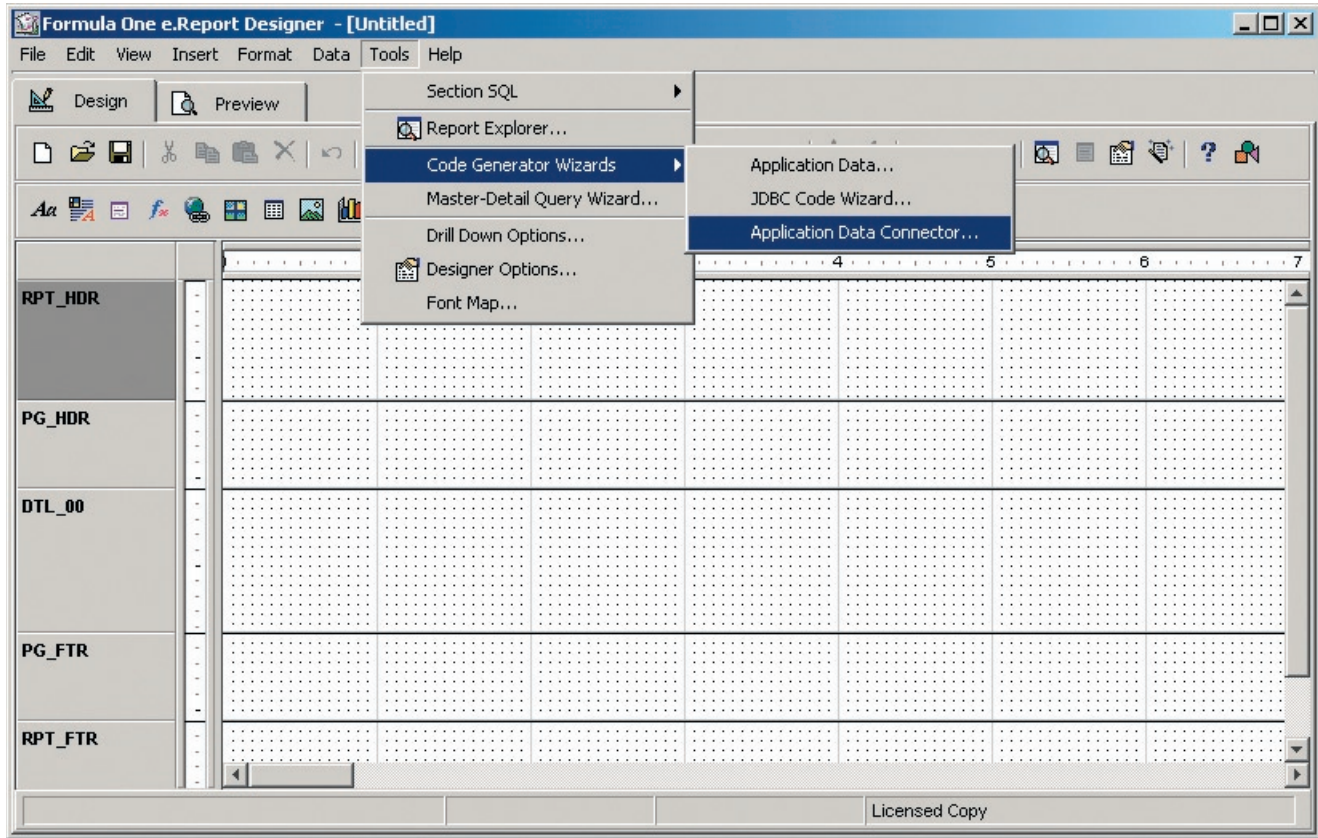
► Scalability and Performance

Small reports are generated quickly using in-memory processing of the relational query that maps to the report sections. Larger reports are also created quickly by leveraging a scalable model where data can be held in secondary storage, memory, or a fine-tuned combination of both.

► Improved Reports

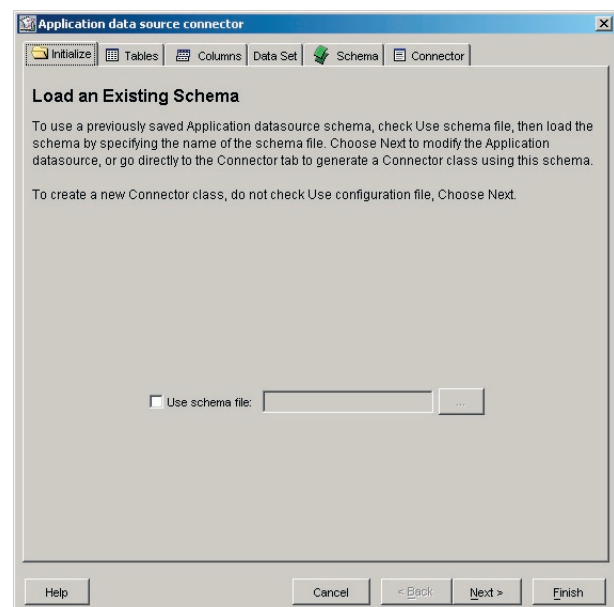
You can access more information easily, which leads to better reports. Objects are manipulated in memory rather than being constrained by the SQL syntax or the limitations of a specific database.





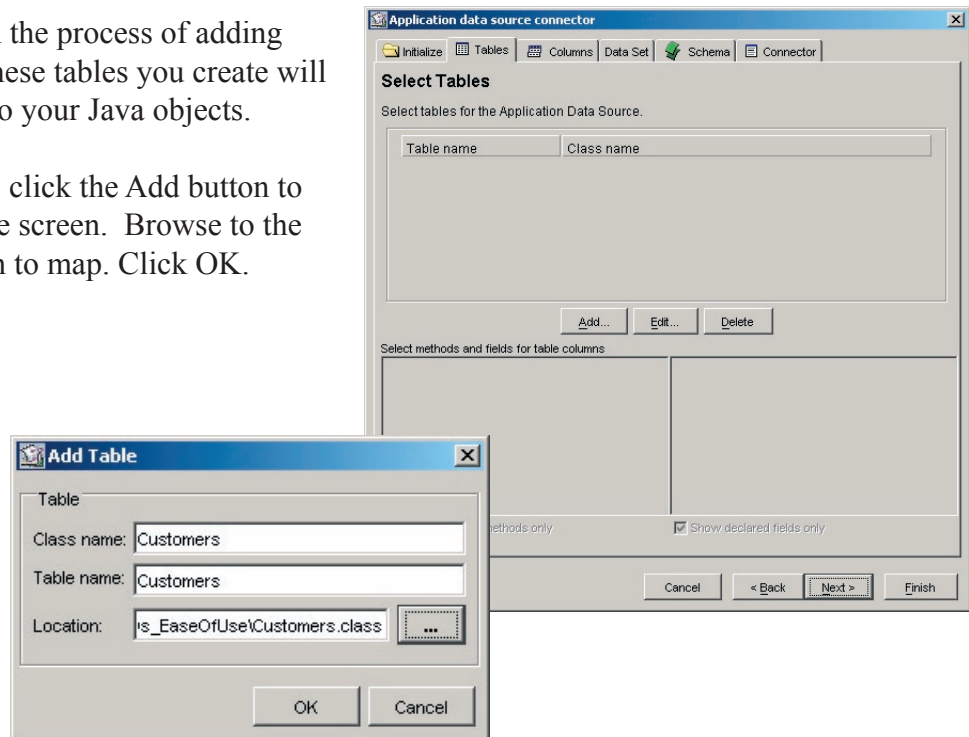
1 First, make sure the Java objects you want to use are in your computer's CLASSPATH so the Formula One e.Report Designer can access them. Next, launch the Designer (above) and select the following menu item: Tools | Code Generator Wizard | Application Data Connector.

2 The Application Data Source Connector dialog box (right) appears. You can load an existing schema if you have already gone through this process and are just looking to make some modifications. If not, select Next.



- 3 Next, you will begin the process of adding tables. Ultimately, these tables you create will become a mapping to your Java objects.

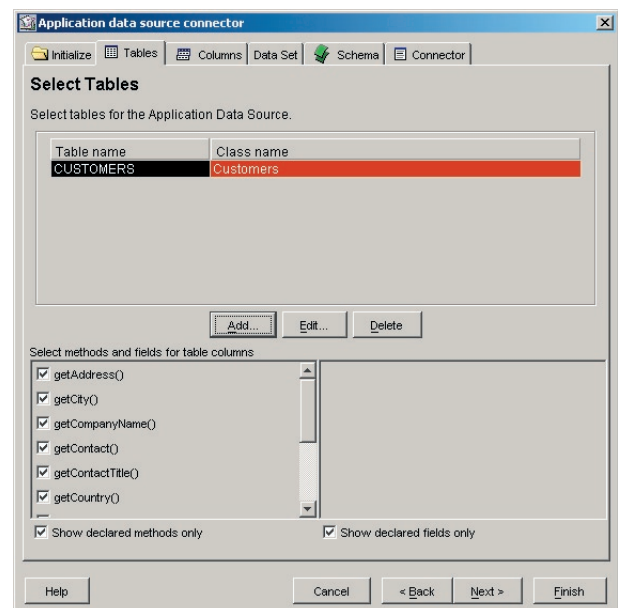
To start this process, click the Add button to launch the Add Table screen. Browse to the Java object you wish to map. Click OK.



- 4 Next, select the Java classes that contain the 'get' methods to retrieve the data from your Java object.

As shown in the previous step, we have selected the Customers class in this example, and it lets me access my customer information.

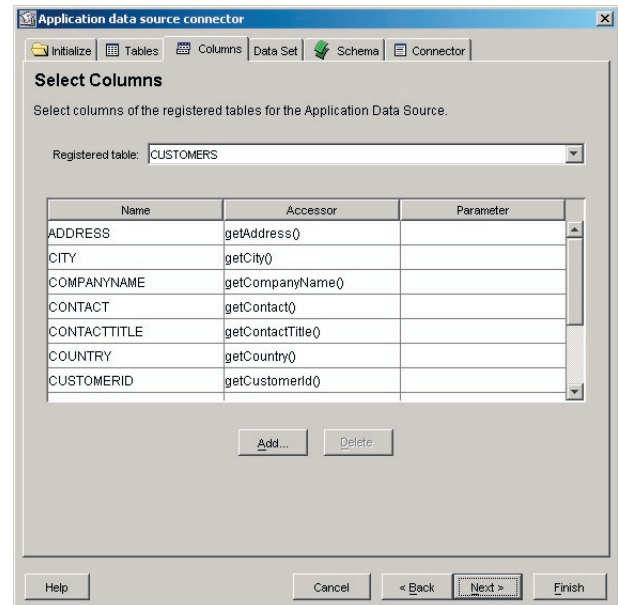
Select the class methods you wish to include in your table and click Next.



- 5** A table named Customers has been created. It has several columns that are connected to your Java object for their data.

Create additional tables from other Java classes by selecting the Tables tab or clicking the Back button.

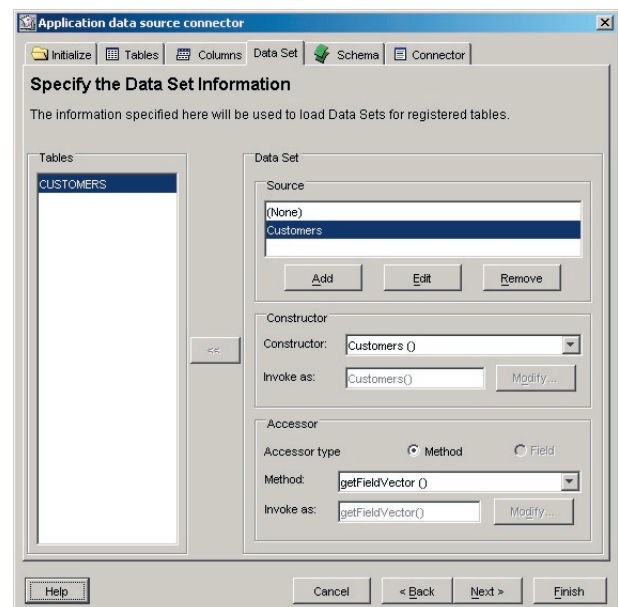
Once you have created all of the tables you need, click Next.



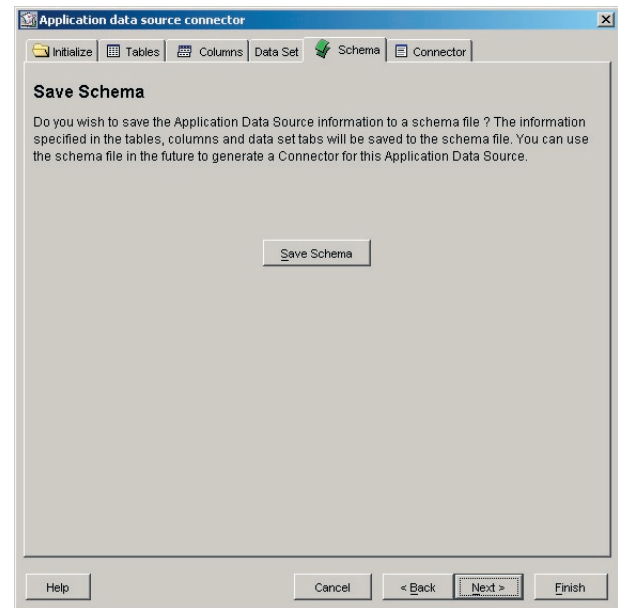
- 6** The Data Set tab is used to associate the newly created table with the location of the data.

In this example, the Customers table will be populated by the Customers class.

Also use this screen to fill out any required information for class constructors.

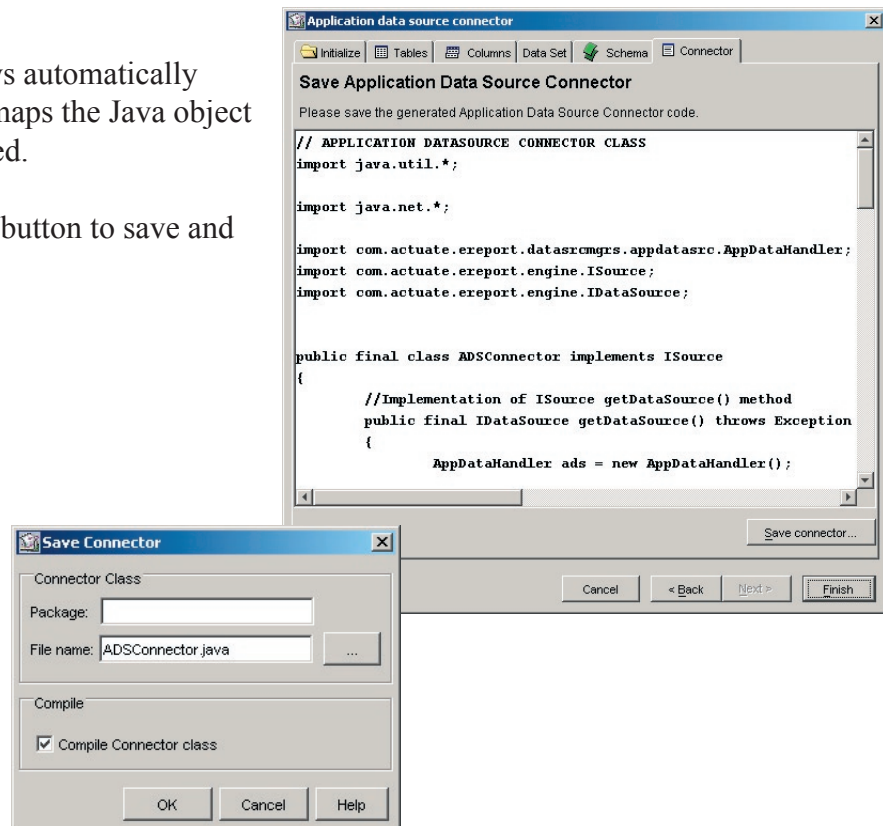


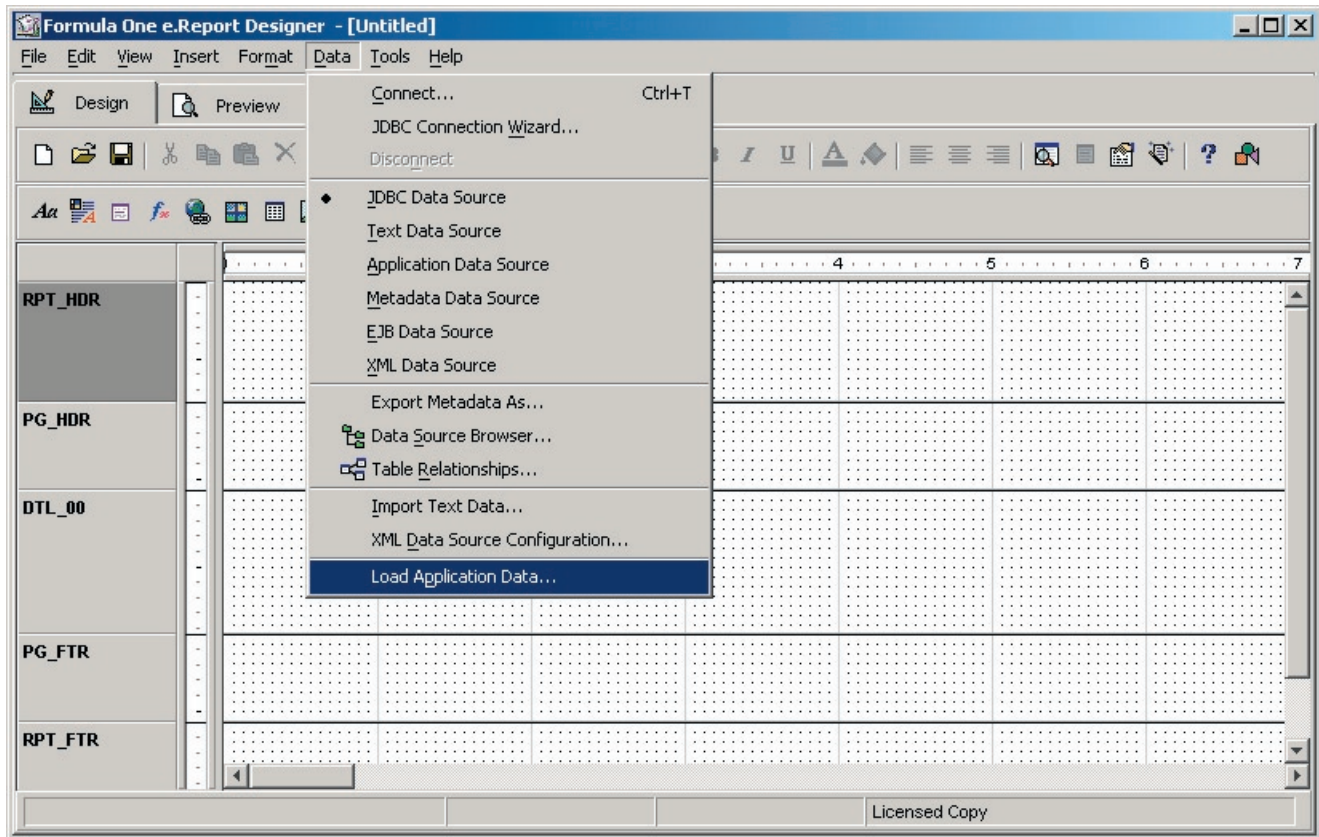
- 7 On the Schema tab, click the Save Schema button to make future modifications without having to recreate the entire set of mappings.



- 8 The Connector tab displays automatically generated Java code that maps the Java object to the table that was created.

Click the Save Connector button to save and compile the code.

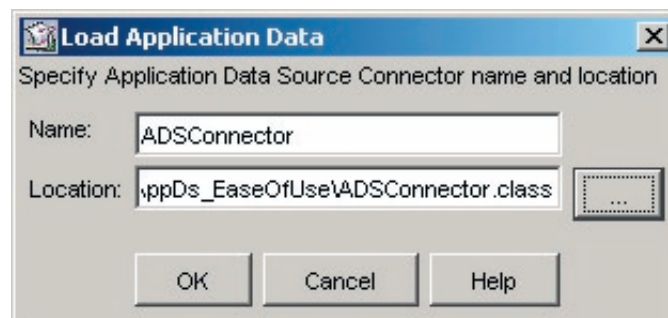




- 9 Now that the object mapping to the table has been completed, it is time to design a report from the connection.

Select Data | Load Application Data from the menu in the Designer.

The screen below appears. Point to your newly created Connector class. (This class was created using the Code Generator wizard from the previous steps.)



- 10** The Java objects now appear as relational tables. This enables you to perform queries and joins as though you were working with database data.

